Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/14/2004
PI Name:	Herman, Cila Ph.D.		
Project Title:	Enhancement of Pool Boiling Heat Transfer and Control of Bubble Motion in Microgravity using Electric Fields-BCOEL		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
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Task Description:	A combined multidisciplinary theoretical and experimental approach is taken to study the heat transfer enhancement in boiling under microgravity. Previous theoretical work will be extended to to accommodate nonuniform electric fields, determine the magnitude of the electric field force components, predict bubble detachment volumes under different experimental conditions, predict bubble trajectories in the liquid and explore possible strategies to control bubble formation, size, coalescence and trajectories. The experimental investigation focuses on visualizing bubble formation, bubble shape, and temperature distribution in the vicinity of the heated surface, using high speed cinematography with and without real-time interferometry. Heat fluxes will be measured and boiling curves will be generated for prescribed heat flux boundary condition at various subcooling levels. The magnitude of the electric field will be varied and different electrode configurations will be used. Bubble injection will be used in the bubble control experiments and force control measurements. Injected bubbles will also be used in the study of bubble trajectories and their control by means of electric fields in 1g and in the KC135 experiments. - Multiple (few) bubbles will be formed by boiling and injection to study the interaction of bubbles, their coalescence with and without electric fields and the influence of coalescence on heat transfer, in 1g lab and in the KC135 experiments. The experimental hardware for the experiments allowing the formation of multiple bubbles by injection has been designed during the first project year. - The behavior of a single vapor bubble, its dynamics and interaction with the thermal boundary layer under various electric field configurations with and without electric fields using holographic interferometry will be studied. - During the first project year experiments were carried out under terrestrial conditions for different bubble injection regimes (wide range of air flow rates and Weeber numbers) to study the influence of the electric field on bubble formation under a variety of conditions. - The analysis of experimental data from low gravity experiments on board of the KC-135 continued. The analysis focused on the influence of the gravity level (microgravity, Lunar gravity, Martian gravity as well as 1g) on bubble detachment under the influence of electric fields. In the nucleate boiling regime, high heat fluxes can be achieved driven by relatively small temperature differences between the solid surface and the bulk fluid. Therefore, pool boiling heat transfer is important in the development of efficient cooling schemes in microgravity applications. The objective of this task is to i) explore the physical mechanisms responsible for the electrohydrodynamic (EHD) heat transfer enhancement in pool boiling in low gravity conditions, ii) quantify the magnitude of enhancement, iii) achieve better control of the boiling process, iv) influence bubble formation, and v) influence bubble motion. Results of the investigations of simple and augmented boiling surfaces will provide valuable insights for the design of phase change equipment for microgravity applications. Understanding of the mechanisms that affect the heat transfer is crucial for practical applications where pool boiling is needed. The effects of introducing the electric field on heat transfer is also crucial from an industrial standpoint. Example of issues this research will help answer are: whether strong electric fields can prevent coalescence of bubbles, and whether the applied fields can improve and control the bubble diameter on detachment. Better understanding of the effect of electric fields on boiling in microgravity and our ability to predict heat transfer rates under these conditions will open new avenues for aerospace technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Key conclusions concerning the influence of electric field in microgravity conditions include 1) Bubble volume decreases and the elongation increases with increasing electric field magnitude, 2) The detachment contact angle changes, 3) Bubbles move away from the electrode at which they have formed 4) Bubble coalescence is less pronounced, 5) Regular bubble detachment is observed in microgravity under the influence of electric fields, contrasted to irregular bubble shape, behavior and detachment without electric fields, 6) The influence of electric fields on gas-liquid two-phase systems is more pronounced in microgravity than in terrestrial conditions. Key conclusions concerning the influence of gravity level (terrestrial, lunar, martian, microgravity) and electric field on bubble formation include 1) The bubble volume at detachment and detachment time decreased considerably as the gravity level increased (from microgravity, through lunar and martian gravity conditions). No appreciable change on bubble shape and formation was observed. 2) An initially uniform electric field elongates the bubble in the direction of the electric field lines. When increasing the applied electric field magnitude bubble volume at detachment decreases, bubble aspect ratio at detachment increases. 3)The effect of the electric field is more pronounced under microgravity conditions than in terrestrial conditions or reduced gravity conditions (lunar, martian) 4) For the investigated electric field magnitude, the impact of electric potential difference on bubble detachment is not as pronounced as the variation of gravity. 5) In terrestrial conditions a nonuniform electric field considerably modifies the bubble shape and trajectory during formation, detachment and ascent into the liquid. 6) The electric field impacts the bubble trajectory in terrestrial conditions: different behavior types were observed.
Bibliography Type:	Description: (Last Updated: 04/13/2009)
Articles in Peer-reviewed Journals	Herman, C., Iacona E., "Modeling of bubble detachment in reduced gravity under the influence of electric fields and experimental verification," Heat and Mass Transfer, Vol. 40, No. 12, pp. 943 – 957. , Jan-2004
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Papers from Meeting Proceedings	Iacona, E., Herman, C., Chang, S., Liu, Z., 2004, "Electric field effect on bubble detachment in reduced gravity environement," Proc. ASME -- ZSIS International Thermal Science Seminar II, Eds.: A. Bergles et al., Bled, Slovenia, June 13-16, 2004, pp. 581-587. Jun-2004
Papers from Meeting Proceedings	Herman, C., Mewes, D., Radke, J., Liu, Z., "Bubble detachment and coalescence under the influence of electric fields," Proc. ASME -- ZSIS International Thermal Science Seminar II, Eds.: A. Bergles et al., Bled, Slovenia, Jun-2004